

SHRUTI CHAITANYA

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SUMMARY

Health informatics professional with 5+ years of combined clinical and healthcare data analytics experience. Master's in health informatics from Indiana University with expertise in healthcare analytics, data visualization, modelling, and data design leveraging SQL, R, and Python. Skilled in bridging clinical workflows with advanced data solutions and EHR optimization to improve patient outcomes, operational efficiency, and data-driven decision-making

EDUCATION

Master of Science Health Informatics- Indiana University, Indianapolis (Luddy School of Informatics) Dec 2025
Coursework: Biomedical Analytics, Clinical Decision Support Systems, Clinical Information Systems, Electronic Health Records, Health Information Exchange

Bachelor of Dental Surgery- Sri Aurobindo College of Dentistry (India) Nov 2021

WORK EXPERIENCE

Clinical Data Analyst Intern – GTechnologies Pvt Ltd Aug 2025 – Dec 2025

- Conducted the **design and development** of an **AI-enabled Electronic Health Record (EHR) system** (MySQL, React.js), improving data transition speed and operational efficiency by **19%**
- Oversaw end-to-end database schema design, deployment, and UI/UX optimization, to achieve a **30% reduction in form completion time** for clinical users
- Integrated predictive analytics (**TensorFlow, Scikit-learn, Keras**) for real-time clinical decision support, enhancing diagnostic accuracy in simulated trials
- Implemented HL7 FHIR compliant data exchange protocols and validation rules, **reducing data integration errors by 25%** to ensure seamless interoperability

Team Lead- BioHealth Informatics Research Center, Indiana University Jan 2025 – Dec 2025

- Administered the organization and cleaning of **200+ public health datasets monthly** for the university data repository, ensuring data accuracy, consistency, and readiness for analytics
- Led and mentored a cross-functional team of **~30** student researchers, overseeing project timelines, data validation protocols, and quality assurance
- Designed **standardized metadata documentation** and implemented streamlined workflows, reducing project turnaround time and saving **60+ hours weekly**

Graduate Research Assistant- BHIRC, Indiana University Aug 2024 - Dec 2024

- Partnered with domain experts to **identify key metrics, define data requirements, and create quality management frameworks** for IU's healthcare analytics data coverage program to ensure technical and clinical accuracy standards
- Built automated data quality assessment reporting for 100+ datasets using Python (Pandas, NumPy, openpyxl), **improving data availability by 200%**
- Enabled data-driven decision-making and research recommendations for public health initiatives by building **interactive Tableau dashboards**

Dentist & Clinical Systems Analyst- Dr Vrinda Kawthekar Clinic Dec 2022 - Nov 2023

- Streamlined EHR workflows by automating the current appointment and record management process, **cutting average consultation time from 45 minutes to ~30 minutes (33% reduction)**
- Developed an analytical framework integrating surgical and appointment data with patient demographics to streamline scheduling workflows and **increase quarterly patient engagement (response rate, attendance rate) by 20%**
- Leveraged CBCT scans, OPG, IOPA, and dental history to design targeted post-surgery care strategies

SKILLS

- Programming:** Python, SQL, MySQLDB, Snowflake, R, XML, JSON, JavaScript, HTML, SAS
- Analytics:** Biostatistics, EDA, Data Visualization, Tableau, Power BI, Machine Learning, Regression, Classification, Hypothesis Testing, A/B Experimentation, Requirement analysis, Predictive Analytics, ETL pipelines, Data governance, KPI tracking, Artificial intelligence
- Healthcare standards & terminologies:** HEDIS, ICD-10, CPT, RxNorm, HL7 FHIR, SNOMED, LOINC, MS-DRG, EHR (Build and configuration/Ticketing/Go-live support/flowsheets), CDA/CCDA, OpenEMR, Epic(Ambulatory/Inpatient/Caboodle), Cerner, OpenMRS, Clinical Documentation, Rest API, SynMedical, PostMan, HIPAA compliance, CMS, HITECH
- Other:** Microsoft 365, Advanced Excel, MS Access, Cross-functional team collaboration, Project management, Agile, Jira, Confluence, Scrum
- Certification:** CITI Biomedical Researcher (Mar 2024), CITI Social/Behavioral Researcher (Jun 2024), HIPAA Privacy and Security (Jan 2025)

PROJECTS

Healthcare Data Normalization and API Integration for Clinical Analytics

- Crafted an **ETL pipeline** in **Python** to extract patient information from **OpenEMR, Hermes Terminology Server**, and clinical repositories via **FHIR-compliant REST APIs**
- Standardized clinical datasets** through industry schema mapping and **developed QA documentation** to improve data accuracy, interoperability, and compliance with FHIR-compatible downstream applications
- Achieved **100% terminology standardization, 40% reduction in manual coding effort**, and improved data accuracy for faster, insight-driven decision-making with real-time analytics support
- Skills used:** SNOMED CT, RxNorm, Pandas, NumPy, RestAPI, JSON/XML Parsing, HL7 FHIR, Data Engineering

Analysis of Vaccine Hesitancy Across U.S. Counties

- Evaluated** vaccine hesitancy patterns across 3,143 U.S. counties, incorporating demographic, socioeconomic, and public health indices
- Performed **exploratory data analysis**, correlation analysis, and predictive modeling to identify statistically significant predictors affecting vaccination rates
- Translated analytical findings into reports and **executive summaries**, developed a **standardized data dictionary**, and recommended intervention strategies
- Skills used:** R, ggplot2, statistical modeling, ANOVA, hypothesis testing, chi-square test, correlation analysis, random forest, lasso regression, feature engineering, shell scripting, Power BI

Maternal Mental Health Analytics During COVID-19

- Conducted **healthcare data analytics** on a cohort of **10,700+** pregnant women, using **predictive modeling** to identify key risk factors for depression.
- Designed an **analytics framework** that transformed raw clinical and demographic data into **data-driven insights** for **population health management** and **evidence-based decision-making** in healthcare
- Developed and compared machine learning models (XGBoost, Random Forest, Logistic Regression, KNN) to predict depression severity, achieving **89% accuracy and AUC = 0.99 for severe cases**
- Skills used:** Python, SQL, Data cleaning, Outlier detection, EDA, Scikit-learn, SMOTE, Model evaluation, Hyperparameter tuning, Confusion matrix